

Week 7 Project Summary

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1. What Was Planned

Test whether the 5 core KPIs and two headline findings (previous no-show history, distance) from Weeks 5-6 are accurate and reliable, test the dashboard's Appointment Type filter, and complete a meaningful cross-track testing activity with the Data Science track.

2. What Was Actually Tested

Independently recalculated all 5 KPIs directly from the raw dataset, simulated the dashboard's Appointment Type filter against all four appointment types, re-validated the previous-no-show finding, and stress-tested the distance finding by filtering it down to a single appointment type to check whether it held up outside the whole-dataset view.

3. Most Important Testing Results

- All 5 KPIs passed independent recalculation with zero discrepancy.
- The dashboard's Appointment Type filter behaves correctly across all four appointment types.
- The previous no-show history finding fully validated with no issues.
- The distance finding validated overall, but a minor non-monotonic dip was found in one intermediate distance band when filtered to a single appointment type.

4. Issues or Weaknesses Identified

A minor, likely sample-driven dip in the 5-10km distance band's no-show rate when the dataset was filtered to Follow-up appointments only, which slightly broke the otherwise steady upward trend seen in the whole-dataset view.

5. Improvements/Refinements Made

Added an explicit caveat to the distance finding's documentation: the relationship between distance and no-show rate is a general upward trend, not a strictly linear one, particularly within smaller filtered subgroups. No KPI or dashboard filter changes were needed, as all passed testing without issue.

6. Retesting Results

After adding the caveat, the underlying finding was re-confirmed to remain valid and useful; the caveat only affects how the finding is described and used, not whether it holds.

7. Track(s) Collaborated With

Data Science.

8. What Was Tested Collaboratively

Whether the analytical findings underlying the Week 6 feature recommendation (previous_no_shows, and a distance x prior-history interaction) remain valid under independent recalculation and additional segment-level stress-testing.

9. What Changed As a Result

The Data Science track now has an explicitly caveated understanding that the distance relationship is a general trend rather than strictly linear, which should inform how that feature is encoded in their model (e.g., continuing to use binned distance ranges rather than a raw linear value).

10. Key Findings / Validation Outcomes

- All Week 5-6 KPIs and headline findings are confirmed accurate and reliable through independent testing.
- The dashboard's filter functionality is confirmed to work correctly.
- The distance-no-show relationship is directionally real but not perfectly linear.

11. Major Challenges

Designing a genuine stress test rather than simply re-running the same calculation again — this required filtering the distance analysis down to a single appointment type to see if the pattern held up in a smaller, more specific subgroup, which is what actually surfaced the minor non-monotonic dip.

12. Important Decisions Made

Decided to document the distance finding's minor irregularity honestly as a caveat, rather than either ignoring it or overstating it as a failed validation, since the core pattern still held directionally.

13. Remaining Limitations

- The 3+ previous no-shows segment remains based on a small sample size (n=93).
- The dataset remains fictional, so findings may not generalise to a real clinic.

14. Remaining Issues or Dependencies

None outstanding. All planned Week 7 tests were completed and either passed cleanly or were resolved with a documented caveat.

15. Contribution to the Overall HealthConnect Project

This week's work confirms that the analytical foundation feeding into the Data Science track's model is accurate, tested, and honestly documented — reducing risk ahead of Week 8 final integration and giving the wider HealthConnect team confidence that the Data Analytics outputs are trustworthy.

16. What Must Be Completed Before Week 8

No outstanding testing work remains for the Data Analytics track. Final polish of the dashboard presentation and ensuring the distance caveat is reflected in any stakeholder-facing summary are the only remaining light-touch items.